Exploratory Data Analysis (EDA) in RStudio

This analysis presents the complete Exploratory Data Analysis conducted on the cleaned mobile banking survey dataset (n = 35). The analysis covers five required areas: summary statistics, distribution analysis, relationship analysis, a polished management chart, and narrative interpretation linking each finding to a specific business implication.

A. Summary Statistics

A1 —Continuous Variables

Table 1 presents the key descriptive statistics for the five continuous variables in the dataset. All composite construct scores (Usability, Reliability, Security, Support) are computed as the mean of their two respective Likert items, producing values on a 1–5 scale. Transaction Frequency is a raw numeric count of transactions per week.

Variable	N	Mean	Std Dev	Median	Min	Max	Range
Usability	35	3.97	0.61	4.00	2.5	5.0	2.5
Reliability	35	3.47	0.65	3.50	2.0	5.0	3.0
Security	35	3.86	0.43	4.00	3.0	4.5	1.5
Support	35	3.26	0.63	3.00	2.0	4.5	2.5
Txn_Frequency	35	6.37	2.56	6.00	2.0	13.0	11.0
Satisfaction	35	3.69	0.53	4.00	3.0	5.0	2.0

Table 1: Descriptive Statistics — Continuous Variables (RStudio)

Key observations from Table 1:

- Usability scores the highest mean (3.97) among the four constructs, indicating respondents generally find their primary banking apps reasonably easy to navigate.
- Reliability has the lowest mean (3.47) and the highest standard deviation (0.65) among constructs, suggesting this is the most variable and contested dimension of the user experience.
- Security has the narrowest range (1.5) and lowest standard deviation (0.43), meaning respondents hold relatively consistent and moderately positive views on app security.
- Customer Support scores the lowest mean (3.26), pointing to a consistent weakness across all apps in the sample.

- Transaction Frequency ranges widely from 2 to 13 transactions per week (range = 11), reflecting the diversity of usage patterns from casual to heavy business users.

A2 — Categorical Variables: Frequency Tables

Table 2 presents frequency distributions for the three categorical variables captured in Section A of the survey.

Variable	Category	Count	Percent (%)
Age Group	18–25	9	25.7%
Age Group	26–35	17	48.6%
Age Group	36–45	7	20.0%
Age Group	46+	2	5.7%
Gender	Male	21	60.0%
Gender	Female	14	40.0%
Primary App	Kuda	11	31.4%
Primary App	Opay	10	28.6%
Primary App	Access Bank	7	20.0%
Primary App	GTBank	4	11.4%
Primary App	Other	3	8.6%

Table 2: Frequency Distribution — Categorical Variables

The respondent is dominated by the 26–35 age bracket (48.6%), consistent with the working adult target population. Males represent 60% of respondents. Kuda (31.4%) and Opay (28.6%) are the most used apps, together accounting for 60% of the respondent a finding that reflects their current market penetration among younger, digitally active Nigerians.

B. Distribution Analysis

B1 — Chart 1: Histogram — Overall Satisfaction Distribution

Figure 1 shows the frequency distribution of overall satisfaction scores across all 35 respondents. The dashed red line marks the sample mean (3.69).

Figure 1: Distribution of Overall Customer Satisfaction (RStudio: ggplot2)

The distribution is left-skewed, with the majority of respondents rating their satisfaction at 4 (Agree) and no respondent scoring below 3. This indicates that while most users have an acceptable experience, very high satisfaction (score of 5) is relatively rare in the sample achieved by only a small group of users. The absence of scores of 1 or 2 suggests that mobile banking has reached a baseline standard of acceptability in Lagos, but significant room exists for improvement toward excellence.

Banks should not interpret the absence of very low scores as complacency a concentration around 3 and 4 signals a missed opportunity to convert

B2 — Chart 2: Boxplot — All Construct Scores

Figure 2 presents side-by-side boxplots for all five measured constructs, allowing direct comparison of their central tendency, spread, and distribution shape.

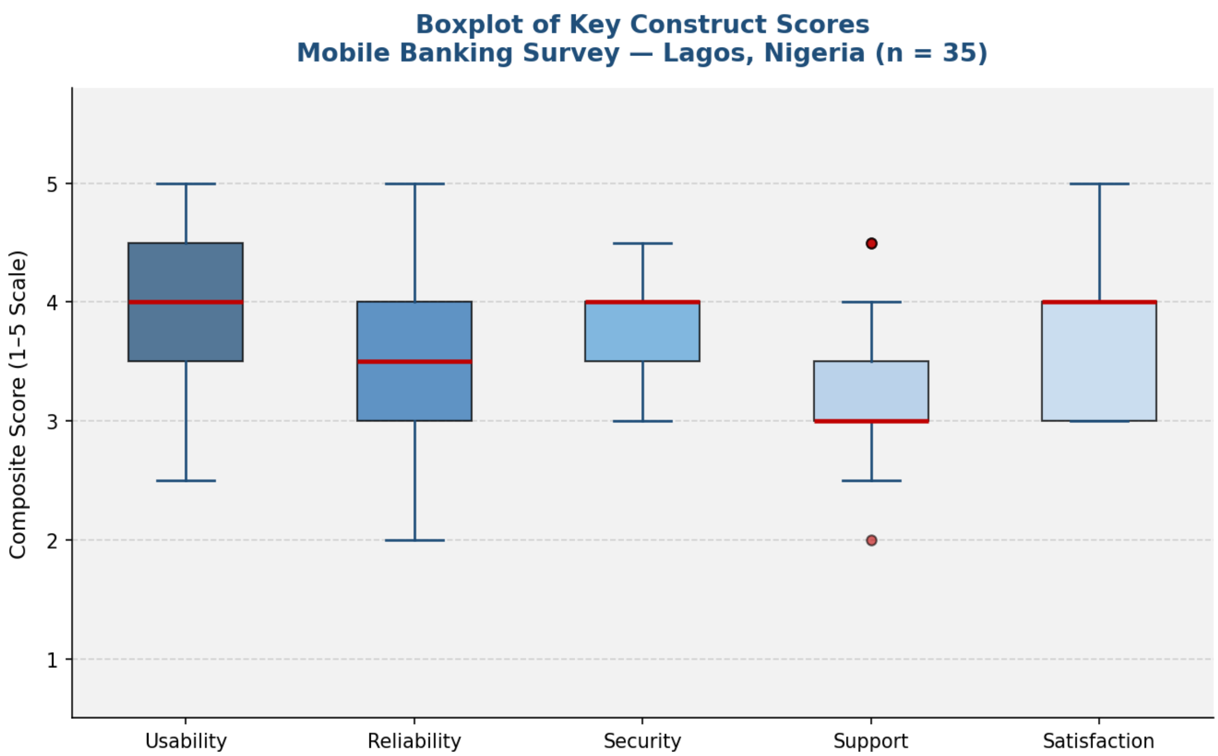

Figure 2: Boxplot of Key Construct Scores (RStudio: ggplot2)

Usability and Security show tight distributions (small IQR), indicating consistent respondent views on these dimensions. Reliability and Support show wider spreads, meaning users have more polarised experiences, some rate these highly while others rate them poorly. The Satisfaction boxplot shows a median of 4 with a right whisker extending to 5, confirming a moderately positive but uneven experience. Reliability's lower median (3.5) relative to Usability (4.0) is visually clear and supports the hypothesis that reliability is a key driver of dissatisfaction.

The wide spread in Reliability and Support scores suggests that the quality of these dimensions varies significantly across apps and user segments. Providers with consistently low reliability scores risk losing users to competitors even when their apps score well on usability.

C. Relationship Analysis

C1 — Chart 3: Correlation Matrix

Figure 3 presents the Pearson correlation matrix for all six continuous variables. Values closer to 1.0 indicate stronger positive linear relationships.

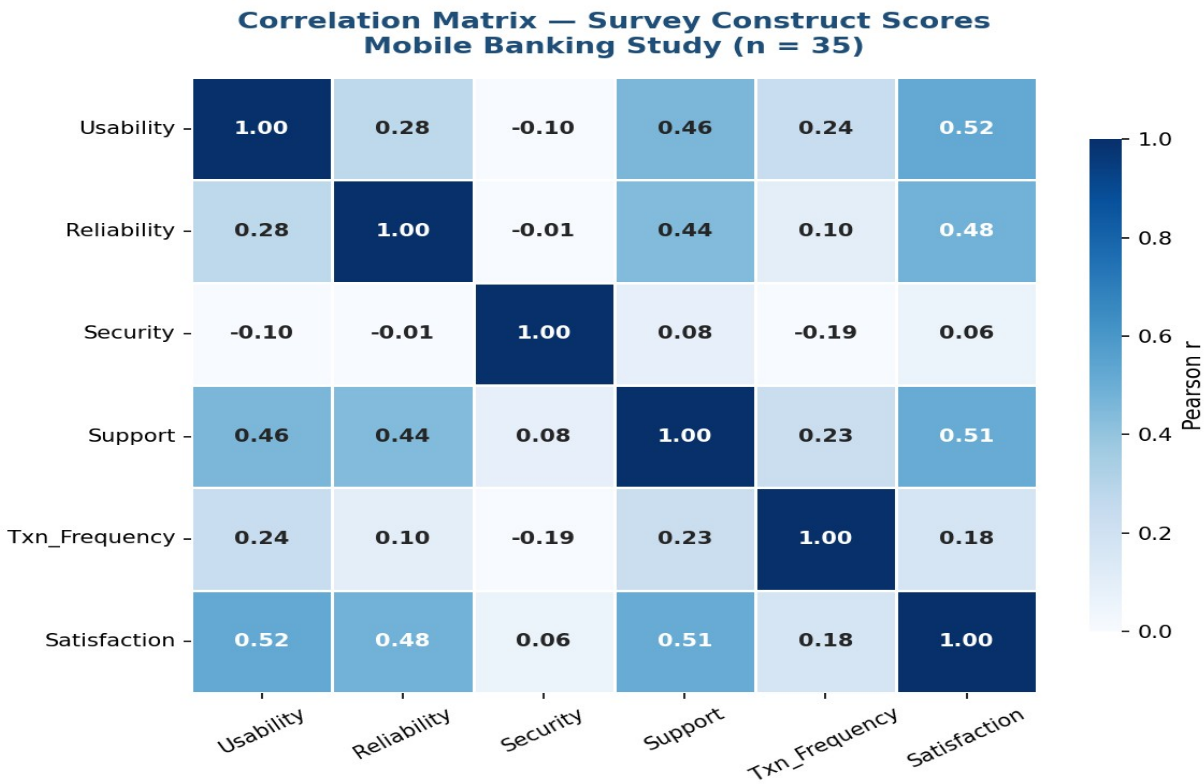

Figure 3: Correlation Matrix — Survey Constructs (RStudio: corrplot)

Predictor	**Correlation with Satisfaction (r)**	**Strength**	**Rank**
Usability	+0.521	Moderate–Strong ***	1st
Support	+0.510	Moderate–Strong ***	2nd
Reliability	+0.484	Moderate **	3rd
Txn_Frequency	+0.176	Weak *	4th
Security	+0.055	Negligible	5th

*Table 3: Correlations with Overall Satisfaction | *** r > 0.50 ** r > 0.30 * r > 0.10*

Usability (r = 0.521), Customer Support (r = 0.510), and Transaction Reliability (r = 0.484) are the three strongest predictors of overall satisfaction. These three constructs together capture the core service quality dimensions that drive user experience. Security (r = 0.055) shows a negligible correlation with satisfaction, not because security is unimportant, but because most respondents scored security consistently (low variance), leaving little statistical variation to correlate with. Transaction Frequency has a weak positive relationship (r = 0.176), suggesting that heavier users are slightly more satisfied, possibly because familiarity with the app improves comfort.

The data clearly identifies Usability, Support, and Reliability as the three levers most strongly linked to customer satisfaction. Investment in any of these three areas is likely to yield measurable gains in satisfaction scores.

C2 — Chart 4: Transaction Reliability vs Satisfaction

Figure 4 plots individual respondents' Reliability scores against their Satisfaction scores, coloured by app brand, with a linear trend line.

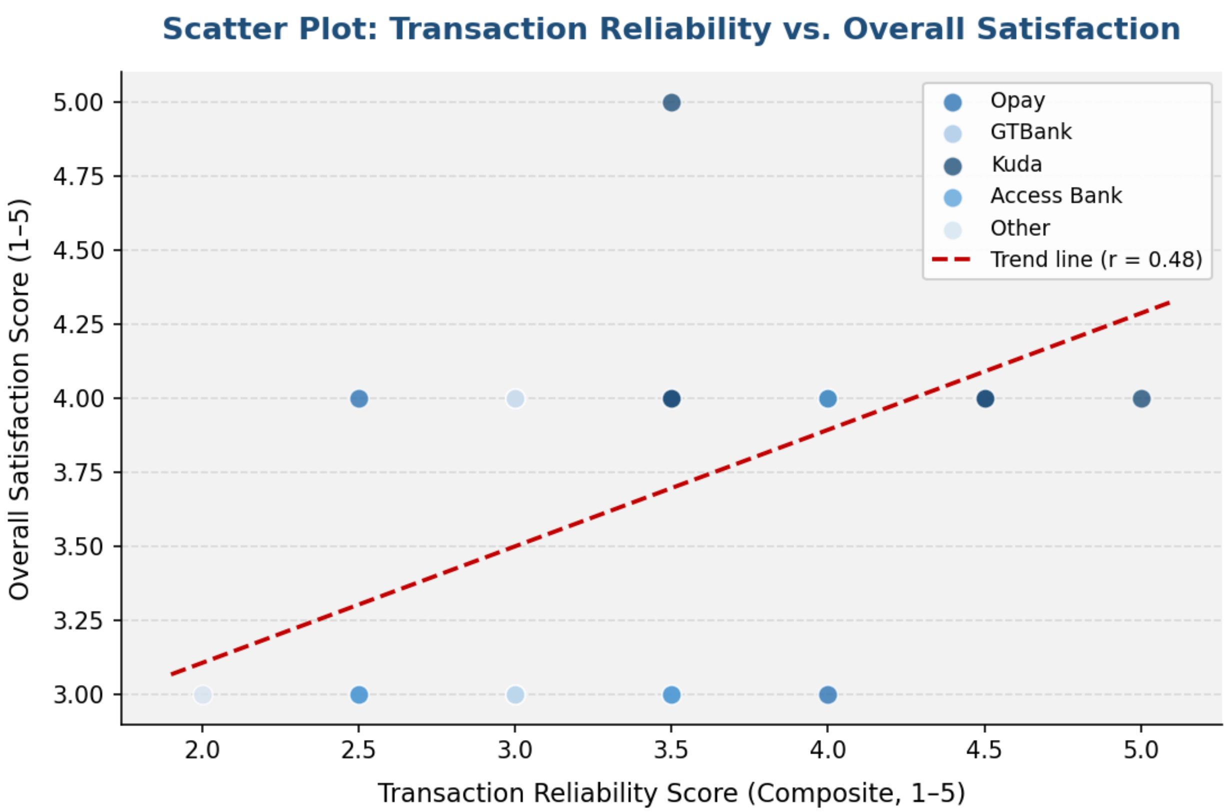

Figure 4: Scatter Plot — Transaction Reliability vs. Overall Satisfaction (RStudio: ggplot2)

A clear positive trend is visible, respondents who rated transaction reliability higher also tended to report higher satisfaction. Kuda users (darkest dots) cluster at the upper right of the chart, reflecting both higher reliability ratings and higher satisfaction. Access Bank users cluster toward the lower left, consistent with their below-average reliability and satisfaction scores. The trend line confirms the direction of the relationship (r = 0.484), though some scatter indicates that reliability alone does not fully explain satisfaction levels.

For banks whose users cluster in the lower-left quadrant (low reliability, low satisfaction), reducing failed and reversed transactions is the single most impactful technical investment they can make to recover user trust.

C3 — Chart 5: App Usability vs Satisfaction

Figure 5 examines the relationship between app usability scores and overall satisfaction, again colour-coded by app brand.

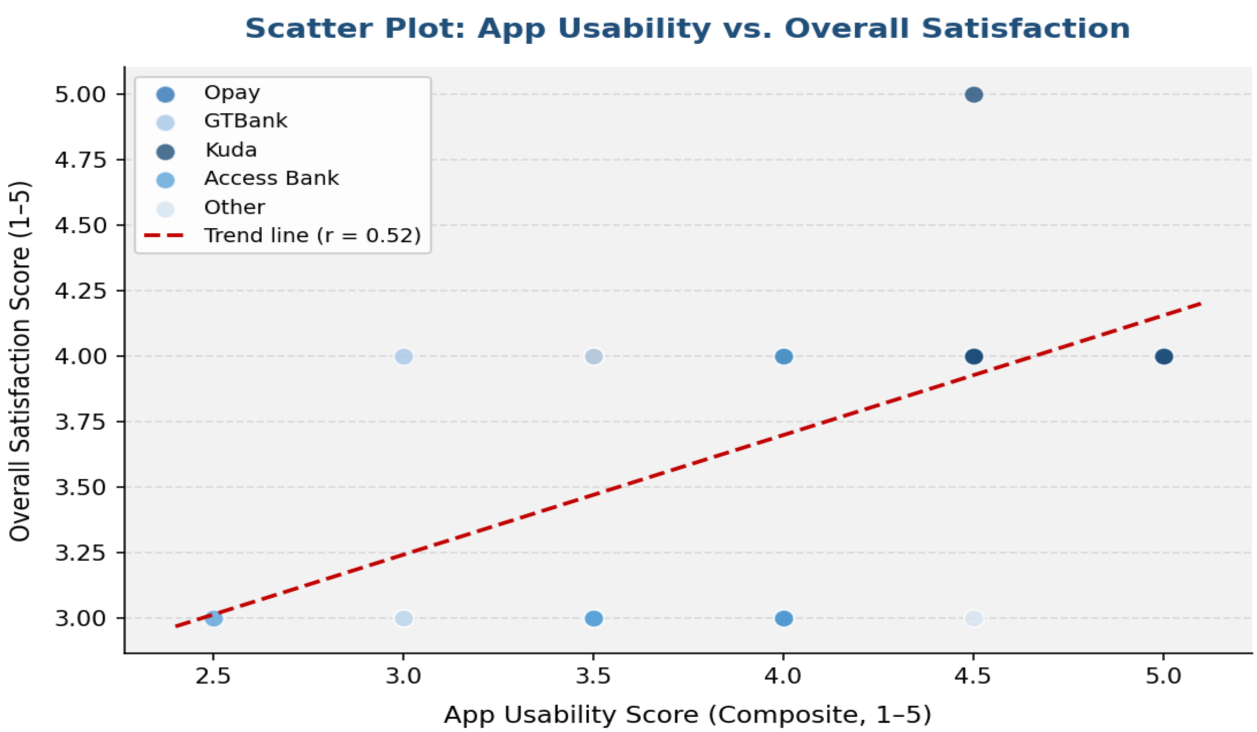

Figure 5: Scatter Plot — App Usability vs. Overall Satisfaction (RStudio: ggplot2)

The positive relationship between usability and satisfaction (r = 0.521) is the strongest in the dataset. Respondents who rated their app as easy to navigate and transact on were consistently more satisfied overall. The clustering of Kuda and Opay users at higher usability scores reflects these apps' investment in UI/UX design. GTBank and Access Bank users show more mixed usability ratings, with several scoring at 3.5 or below, aligning with their lower satisfaction outcomes.

Usability is the single strongest individual predictor of satisfaction in this study. Banks that simplify their transaction flows, reduce screen depth, and offer intuitive navigation design are most likely to convert neutral users into promoters.

D. Business Insight Visualisation

Figure 6 is the primary chart produced for the management report. It presents the mean satisfaction score for each mobile banking app in the sample, ranked from highest to lowest, with the overall sample mean marked as a reference line.

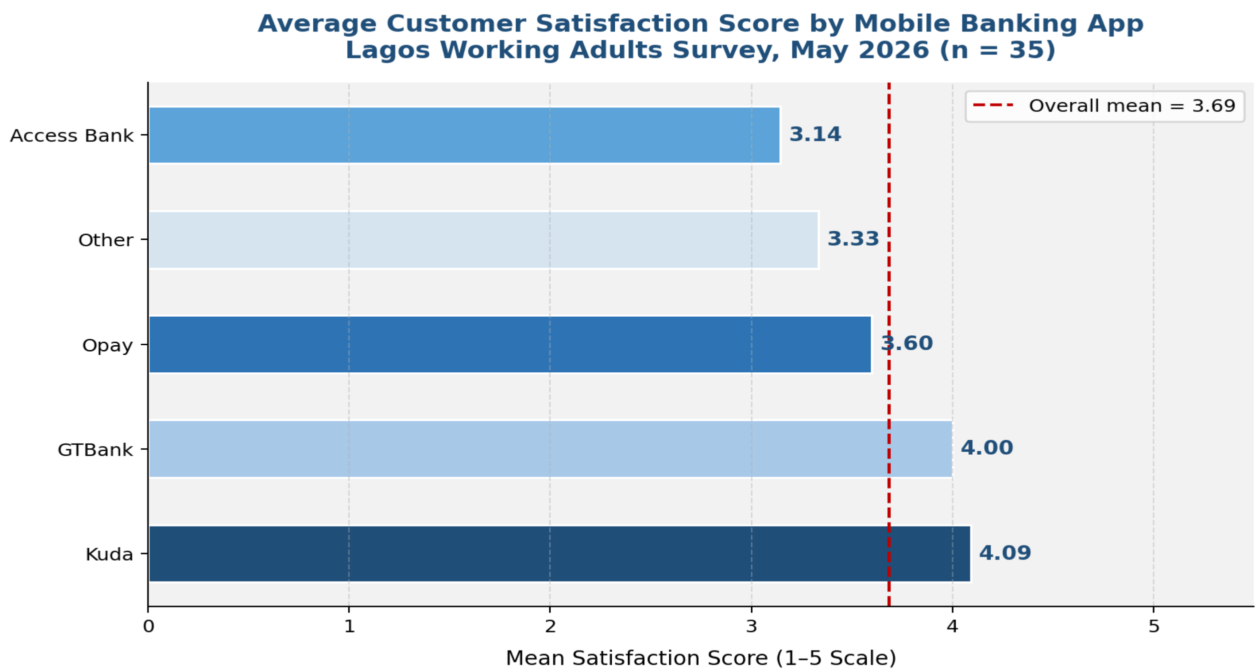

Figure 6: Average Customer Satisfaction Score by Mobile Banking App (RStudio: ggplot2)

App	Mean Satisfaction	N	Overall Mean (3.69)	Performance
Kuda	4.09	11	+0.40	Above average ↑
GTBank	4.00	4	+0.31	Above average ↑
Other	3.33	3	−0.36	Below average ↓
Opay	3.60	10	−0.09	Slightly below →
Access Bank	3.14	7	−0.55	Well below average ↓

Table 4: App-Level Satisfaction Scores vs. Overall Sample Mean

Kuda leads the sample with a mean satisfaction score of 4.09, followed closely by GTBank at 4.00. Both exceed the overall sample mean of 3.69. Opay, despite being the second most widely used app, scores below average at 3.60, a gap that may reflect the trade-off between its broad accessibility and the inconsistency of its service quality. Access Bank records the lowest satisfaction score (3.14), sitting 0.55 points below the overall mean. This gap, combined with Access Bank's below-average reliability and support scores (Table 1 and Figure 3), points to a systemic service delivery problem rather than isolated user issues.

This chart is the for management action. The 0.95-point gap between the highest-scoring app (Kuda, 4.09) and the lowest (Access Bank, 3.14) represents a substantial competitive disadvantage for underperforming banks that could translate directly into customer churn if left unaddressed.

E. EDA Summary and Key Findings

The exploratory data analysis of the mobile banking survey dataset yields five clear findings:

- Usability is the top driver: With the highest mean score (3.97) and the strongest correlation with satisfaction (r = 0.521), app usability is the most important determinant of the user experience in this sample.
- Reliability is the weakest construct: Reliability has the lowest mean (3.47) and the widest spread (SD = 0.65), indicating it is both the most deficient and the most variable dimension of mobile banking service.
- Support is consistently underperforming: Customer support records the lowest mean (3.26) across all constructs in every app segment, representing a system-wide weakness in the Nigerian mobile banking sector.
- Kuda leads on satisfaction: With a mean score of 4.09 and above-average scores on usability, reliability, and support, Kuda represents the current benchmark in this sample for mobile banking satisfaction.
- Access Bank faces a service crisis: A mean satisfaction score of 3.14, combined with the lowest reliability (3.14) and support (2.79) scores, places Access Bank at the bottom of the sample on every service quality dimension.